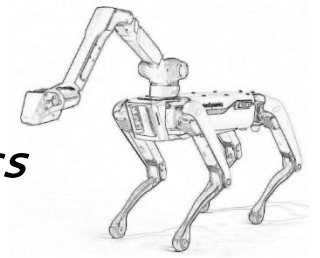


EE 587 Introduction to Robotics

Homework #1



General Info

- The homeworks are for individual study and are for gaining a better understanding of the course topics. Submit a homework that is your own effort.
- In general, use abbreviations for trigonometric functions. i.e., $s\theta = \sin(\theta)$, $c\theta = \cos(\theta)$ etc.
- If you give the coordinates of a vector or a point as an answer, make sure you specify in which frame it is represented. E.g. p^A specified the point resolved (represented) in frame $\{A\}$.
- The programming questions will use Python3 and the “Robotics Toolbox for Python” by P.Corke. This toolbox also uses a sub-toolbox called “Spatial Math Toolbox” that defines the $SO(2)$, $SE(2)$, $SO(3)$ and $SE(3)$ data types with many utility functions. The links to these toolboxes are in oduclass.
- In analytic calculations, leave the square roots in analytic form rather than calculating their decimal values.

Questions

1. Frame $\{A\}$ and Frame $\{B\}$ are initially coincident. Frame $\{B\}$ is rotated initially around unit vector $\hat{x}^B$, by angle θ and then around $\hat{z}^B$ by angle α . Calculate the 3×3 rotation matrix R_B^A that describes the orientation of frame $\{B\}$ with respect to the frame $\{A\}$.
2. Suppose we have $\theta = 45^\circ$ and $\alpha = 30^\circ$. A point is given in Frame $\{B\}$ with coordinate vector $p^B = [3 \ 4 \ 5]^T$. Analytically calculate its coordinate representation in Frame $\{A\}$.
3. Suppose the same coordinates were given for a point q^A in Frame $\{A\}$. Calculate the coordinates of this point in frame $\{B\}$. How did you use the result in (1) to make this calculation?
4. Now again, consider Frames $\{A\}$ and $\{B\}$ coincident. Frame $\{B\}$ is now rotated initially around unit vector $\hat{x}^A$, by angle θ and then around $\hat{z}^A$ by angle α . Again calculate the 3×3 rotation matrix R_B^A that describes the orientation of frame $\{B\}$ with respect to the frame $\{A\}$. Does your result coincide with the result in (1)? Explain.
5. Consider the basic rotations described in (1) with θ and α values given in (2). Create the corresponding basic rotation matrices using the two alternative toolbox functions, e.g., $SO(3).Rx()$ and $rotx()$ with the proper axes specified. Use the computer to calculate the overall rotation matrix R_B^A . (Note: How do you multiply matrices with Python3 and `numpy` package?) Verify your results in (2) and (3).

6. Identify (how in python3?) the data types of the rotation matrices obtained with the two alternative versions of creating the rotation matrices.
7. Explore the options of your chosen plotting function `SO(3).plot()` or `trplot()` such as specifying the Frame label, color, coordinate range etc. Now plot
 - a. The original frame {A},
 - b. The intermediate frames {A₁} and {A₂} resulting from the two basic rotations,
 - c. The final Frame {B}.
8. Use `tranimate()` to animate the overall rotation from Frame {A} to Frame {B}. (There is also a `SO(3).animate()` version).
9. Given the Homogeneous transformation matrix

$$T_A^B = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & \cos(\theta) & -\sin(\theta) & 2 \\ 0 & \sin(\theta) & \cos(\theta) & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- a. Analytically find the homogeneous transformation T_B^A .
 - b. Given $\theta = 45^\circ$ and a point $p^B = [3 \ 4 \ 5]^T$ in frame {B}, compute the coordinates p^A in Frame {A}.
10. Show that $\in SO(3)$, namely it is a Rotation Matrix.

$$R = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{bmatrix}$$

Programming Task:

11. Animate a rotating cube:
 - a. Write a function to plot the edges of a cube centered at the origin. (Note: You will need the `mplot3d` extensions to the `matplotlib` package in Python3)
 - b. Modify the function to accept an argument which is a homogeneous transformation that is applied to the cube vertices before plotting.
 - c. Animate rotation about the x-axis.
 - d. Animate rotation about all axes.

Good luck.